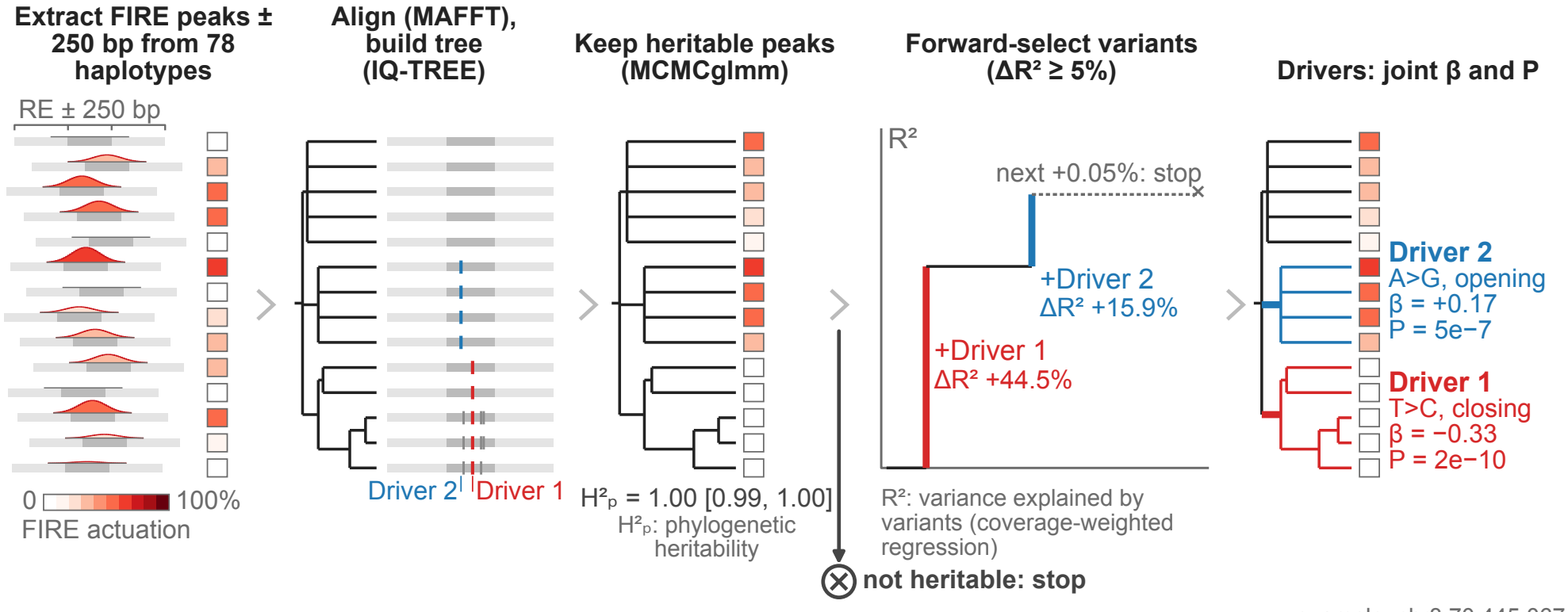

Extract FIRE peaks ± 250 bp from 78 haplotypes
RE ± 250 bp
0
100%
FIRE actuation
Align (MAFFT), build tree (IQ-TREE)
Driver 2
Driver 1
Keep heritable peaks (MCMCglmm)
H²p = 1.00 [0.99, 1.00]
H²p: phylogenetic heritability
not heritable: stop
Forward-select variants (ΔR² ≥ 5%)
R²
next +0.05%: stop
+Driver 2
ΔR² +15.9%
+Driver 1
ΔR² +44.5%
R²: variance explained by variants (coverage-weighted regression)
Drivers: joint β and P
Driver 2
A>G, opening
β = +0.17
P = 5e−7
Driver 1
T>C, closing
β = −0.33
P = 2e−10
example: chr8:70,445,067